

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

EMBEDDINGS OF SIMILARITY ASSESSMENTS

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Partial Information about Distance/Similarity Function

Want an embedding of $x_1, \dots, x_n$ that respects some distance function $d(x, x')$ or similarity function $s(x, x')$. But these functions are unknown!

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Partial information is available in the form of comparisons, e.g.:

- 1 A collection of "similar" pairs and "dissimilar" pairs
- 2 Triplet comparisons: x is more similar to x' than to x''
- 3 Quadruplet comparisons: x is more similar to x' than u is to u'

How to embed based on these?

Example: Triplets

- Input: a collection T of triplets (i, j, k) , meaning that x_i is more like x_j than x_k
- Desired output: Embedding $y_1, \dots, y_n \in \mathbb{R}^k$

